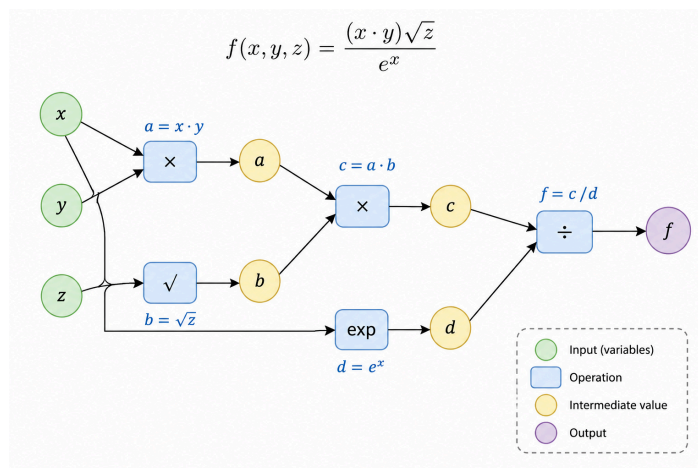


Exercise 3.1: Computational Graphs optional Draw the computational graph for the function:



Using this graph we can compute:

- $\partial f / \partial x$
- $\partial f / \partial y$
- $\partial f / \partial z$

because the graph stores all intermediate operations and dependencies between variables.

Exercise 3.2: Backpropagation. Explain the purpose of the backpropagation algorithm.

Prediction $\rightarrow$ Loss $\rightarrow$ Backward gradient computation

The purpose of the backpropagation algorithm is to compute gradients of the loss function with respect to model parameters. It propagates the prediction error backward through the computational graph so that the weights can be updated to minimize the loss during training.

Exercise 3.3: Gradient Descent Explain the idea behind the gradient descent algorithm. Write down the update formula. What would happen if the sign was inverted?

Gradient descent is an optimization algorithm used to minimize the loss function by updating model parameters step by step in the opposite direction of the gradient.

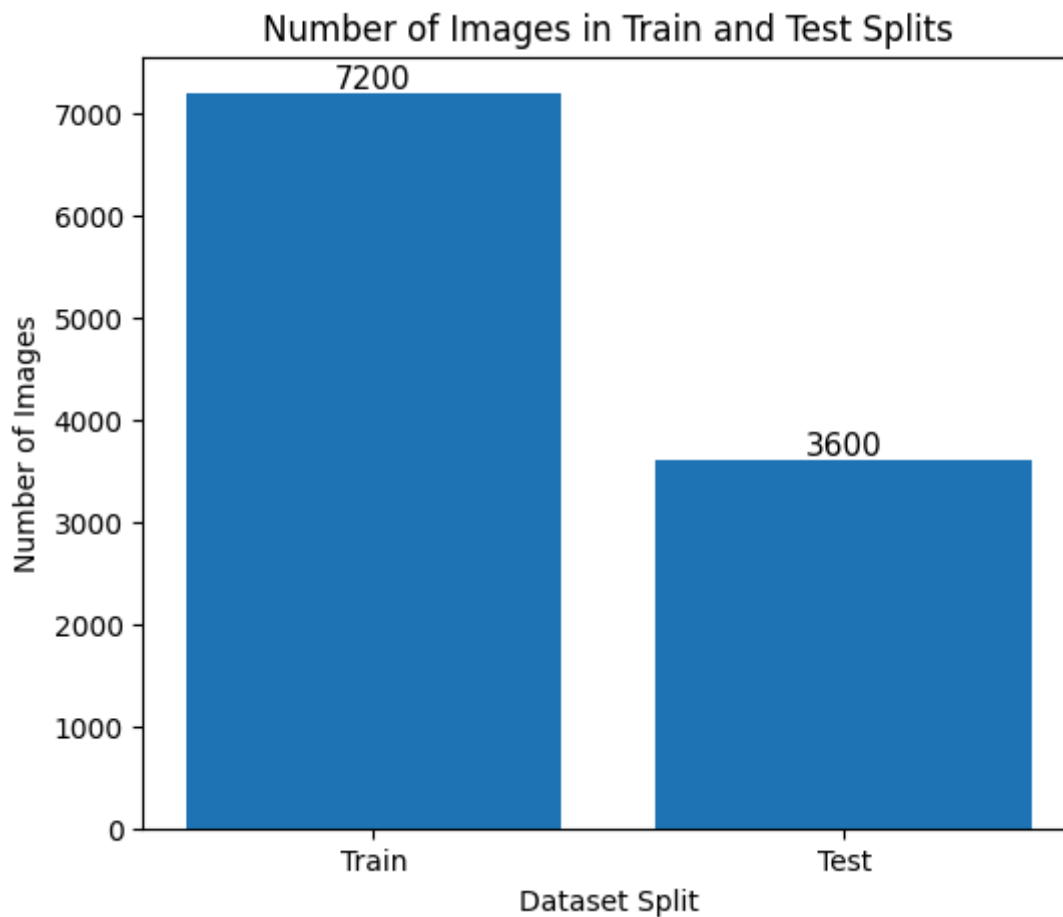
Update formula:

$$\text{Weight}(\text{new}) = \text{Weight}(\text{old}) - (\mathbf{L} \cdot \mathbf{R}) \cdot \text{Gradient}$$

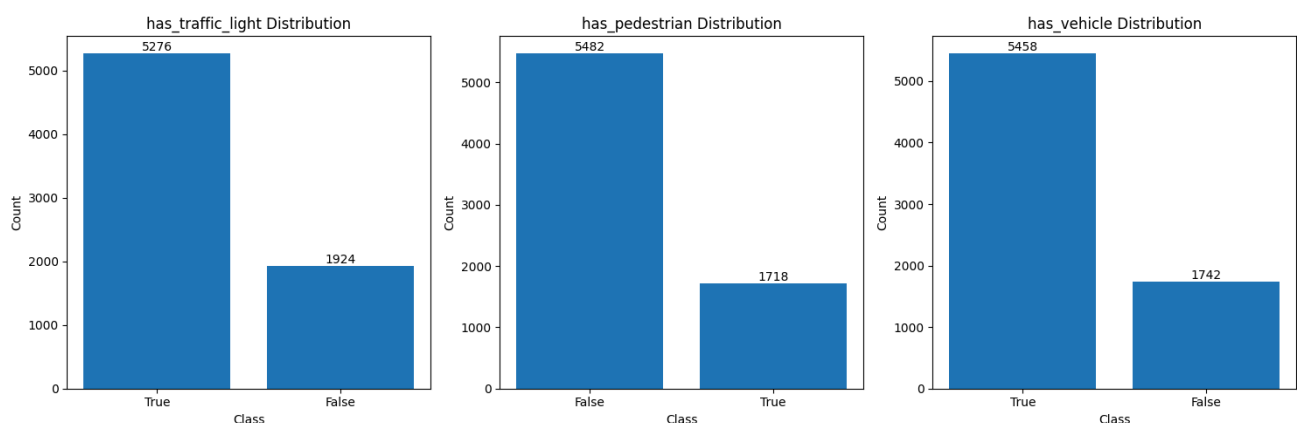
If the sign is inverted, the algorithm would move in the direction of increasing loss instead of decreasing it, causing the model to diverge rather than learn properly.

Exercise 3.4: Dataset Exploration Load the dataset and explore it.

1. How many images are in the training and test splits?



2. What is the class distribution for each label? Are the classes balanced?



The class distribution for the three labels was analyzed using bar charts.

- The **traffic light** label contains 5276 TRUE samples and 1924 FALSE samples.
- The **pedestrian** label contains 1718 TRUE samples and 5482 FALSE samples.
- The **vehicle** label contains 5458 TRUE samples and 1742 FALSE samples.

The classes are not fully balanced.

The pedestrian label is highly imbalanced because FALSE samples are much higher than TRUE samples. The traffic light and vehicle labels also contain more TRUE samples than FALSE samples. Such imbalance may affect model learning and can bias predictions toward the majority class.

3) Display a few example images for each label combination. What patterns do you notice?



A few example images were displayed for traffic light, pedestrian, and vehicle labels to understand the dataset visually.

- In the **traffic light** image, the road scene contains urban buildings and roadside infrastructure, while the traffic light appears as a small object in the distance.
- In the **pedestrian** image, pedestrians appear near the road and sidewalks, but they are relatively small and far from the camera, making detection more challenging.
- In the **vehicle** image, vehicles are visible in front of the ego vehicle along the driving lane and are easier to observe compared to pedestrians.

General patterns noticed:

- Most images represent urban daytime driving scenarios.
- The weather and lighting conditions appear clear and consistent.
- Objects farther from the camera occupy smaller image regions, which may affect detection performance.

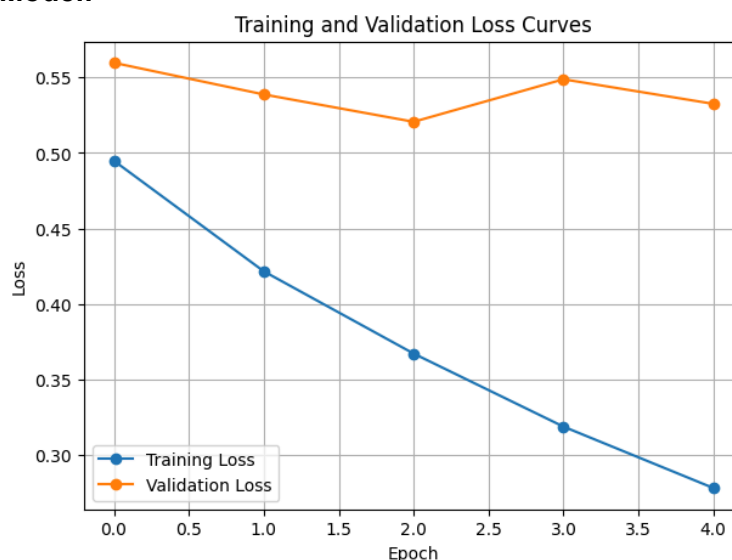
Exercise 3.5: Train Three Binary Classifiers Train three separate binary classifiers on the training split – one for each detection task (pedestrian, traffic light, vehicle). Use Python and preferably PyTorch. You may use a pre-trained backbone (e.g., ResNet-18) and fine-tune it for binary classification. Commit your code to your git repository.

1. Describe your model architecture and training setup (optimizer, learning rate, loss function).

- Used a pre-trained ResNet-18 model for all three classification tasks (pedestrian, traffic light, and vehicle detection).
- Applied transfer learning by modifying the final layer for binary classification output.
- Resized all input images to 224×224 before training the model.
- Converted the images into tensors so that they can be processed by PyTorch.
- Used BCEWithLogitsLoss as the loss function since each task predicts only TRUE or FALSE.
- Used the Adam optimizer with a learning rate of 0.001 for updating model weights.
- Training was done using mini-batches with shuffled data to improve learning performance.

2. Train each model and plot the training and validation loss curves. Do the models converge?

For Pedestrian Model:

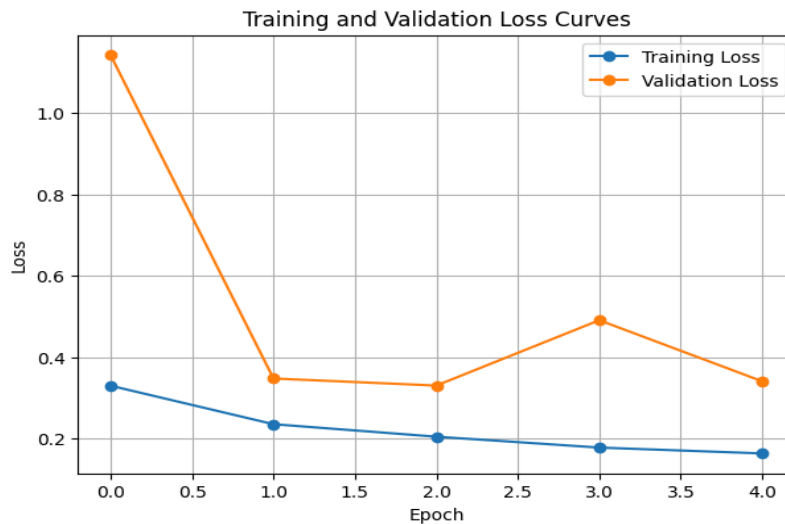


The models were trained for 5 epochs using the training split, while the test split was used for validation. Training and validation loss curves were plotted to analyze the learning behavior of the models.

The training loss decreased steadily across epochs, showing that the model successfully learned patterns from the dataset. The validation loss also remained stable without large fluctuations, indicating reasonable generalization to unseen data.

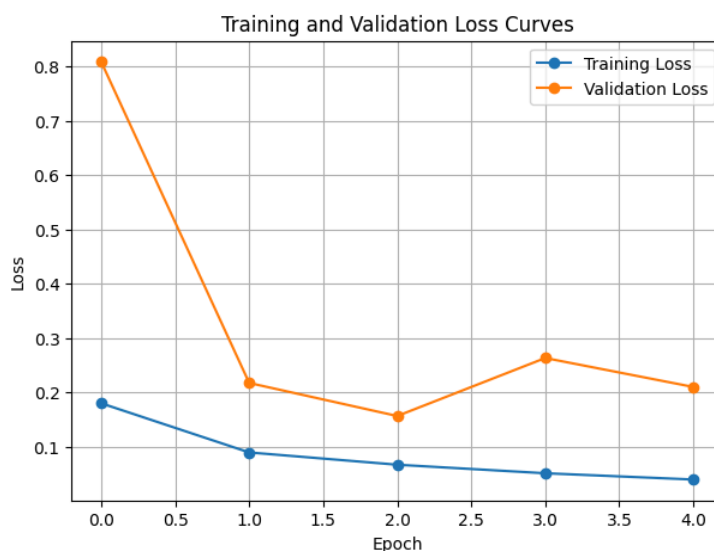
Overall, the model showed convergence because both training and validation losses reduced and stabilized during training. However, the validation loss remained slightly higher than the training loss, suggesting **minor overfitting** on the training data.

For Vehicle Prediction:



- The vehicle detection model was trained for 5 epochs, and both training and validation loss curves were analyzed to study the learning behavior of the model.
- The training loss continuously decreased across epochs, showing that the model successfully learned patterns related to vehicle detection from the training data. The validation loss also decreased significantly after the first epoch and remained relatively stable during later epochs.
- A small increase in validation loss was observed around epoch 4, which may indicate slight overfitting or variation in unseen validation samples. However, the validation loss reduced again in the final epoch.
- Overall, the model showed good convergence because both training and validation losses generally decreased and stabilized during training. This indicates that the model learned meaningful features and was able to generalize reasonably well on unseen data.

For Traffic Light Prediction:



- The traffic light detection model was trained for 5 epochs, and both training and validation loss curves were analyzed to understand the learning behavior of the model.
- The training loss continuously decreased across epochs, showing that the model successfully learned traffic light related features from the training data. The validation loss also decreased significantly after the first epoch and remained relatively stable overall.

- A slight increase in validation loss was observed around epoch 4, which may indicate minor overfitting or variations in unseen validation samples. However, the validation loss reduced again in the final epoch.
- Overall, the model showed good convergence because both training and validation losses generally decreased and stabilized during training. This indicates that the model was able to learn meaningful traffic light patterns and generalize reasonably well to unseen images.

3) Why might training three separate models be preferable to a single multi-label classifier from a safety perspective?

- Training three separate models allows each model to focus on learning one specific task clearly, such as pedestrian, vehicle, or traffic light detection.
- If one model fails or gives incorrect predictions, the other models can still work independently, improving overall system safety.
- Different objects may require different learning patterns and sensitivities. For example, pedestrian detection may need higher recall compared to traffic light detection.
- Separate models make debugging and error analysis easier because the failure source can be identified more clearly.
- In safety-critical systems like autonomous driving, isolating tasks reduces the risk of one classification problem affecting all other predictions in the system.

Exercise 3.6: Evaluation

Evaluate each of your three trained models on the test split.

1. Report accuracy, precision, recall, and F1-score for each model.

	Model	Accuracy	Precision	Recall	F1-score
0	Pedestrian	0.818889	0.600746	0.228045	0.330595
1	Vehicle	0.861667	0.979948	0.832593	0.900280
2	Traffic Light	0.936389	0.939202	0.974458	0.956505

2. Which model performs worst? Hypothesize why.

The **pedestrian detection** model performs the worst compared to the other two models.

Possible reasons:

- Pedestrians appear smaller and farther from the camera.
- Pedestrians may be partially hidden or difficult to distinguish from the background.
- The dataset may contain fewer pedestrian examples compared to vehicles and traffic lights.
- Pedestrian detection is generally more complex because human shapes and positions vary significantly.

3. From a safety perspective, which metric matters most for each model?

- **Pedestrian Detection → Recall is most important**

Missing a pedestrian can lead to serious accidents. It is safer for the model to detect as many pedestrians as possible, even if some false positives occur.

- **Vehicle Detection → Recall is important**

Failing to detect nearby vehicles may cause collisions. High recall helps ensure surrounding vehicles are identified correctly.

- **Traffic Light Detection → Precision and Recall both matter**

The system should correctly identify traffic lights while avoiding incorrect predictions. Wrong traffic light detection may lead to unsafe driving decisions such as incorrect stopping or acceleration.

Exercise 3.7: Comparing the ODD to the Training Data In Exercise Sheet 2 you defined an Operational Design Domain (ODD) based on the system description – before seeing any data. Now that you have explored the dataset and trained your models, revisit that ODD.

1. Look at the data you trained on. What conditions does it cover? List the environmental conditions represented in the training set (weather, lighting, scene types, etc.).

After exploring the dataset and training the models, the following environmental conditions were observed in the training data:

- Mostly clear weather conditions with good visibility
- Bright daytime lighting conditions
- Urban road scenes and intersections
- Presence of vehicles, pedestrians, sidewalks, buildings, and traffic lights
- Low to moderate traffic scenarios
- Front camera driving perspective from city roads

These observations are also visible from the sample RGB images explored during training.

2. Compare this to the ODD you defined in Exercise Sheet 2. Which ODD dimensions are not covered by the training data?

Comparing the dataset with the previously defined ODD, several conditions are not properly represented:

- Night-time or low-light driving conditions
- Heavy rain, fog, or snow weather conditions
- Rural roads, highways, and off-road environments
- Very high-speed driving scenarios
- Unusual pedestrian types or rare human poses
- Complex or highly crowded traffic environments

3. What are the implications of this gap for your models' safety? For which ODD conditions can you not make any claims about model performance?

- The models may fail to generalize correctly in unseen conditions such as rain or night driving.
- Pedestrian detection performance may be reduced in low visibility or crowded scenes.
- Traffic light and vehicle detection may become unreliable in unfamiliar environments like highways or rural roads.
- Since these conditions were not present in the training data, no strong claims can be made about model performance outside the observed ODD.
- Operating the system outside these covered conditions may increase the risk of unsafe decisions and accidents.

This also relates to the ODD safety concern identified earlier, where operating outside the defined ODD was considered hazardous

4. Note down the gaps you identified (if any). You will revisit them in Sheets 4 and 5 when you formally measure ODD coverage and refine the ODD.

The following ODD gaps were identified:

- Lack of adverse weather conditions
- Lack of night-time driving data
- Limited scene diversity outside urban roads
- Limited representation of unusual pedestrian scenarios
- Lack of high-speed driving conditions
- Limited environmental and traffic complexity